



DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY
UNIVERSITY EXAMINATIONS 2025/2026
THIRD YEAR FIRST SEMESTER EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE IN MECHATRONIC ENGINEERING

EMT 3101: ENGINEERING DESIGN PROCESS

DATE: 5TH AUGUST 2025

TIME: 08.30-10.30 PM

Instructions:

This paper contains **FIVE** questions.

Question ONE is compulsory.

Answer **Question One**, **ONE** question in **Section A** and **ONE** question in **Section B**

Question One carries 30 marks, the rest carry 20 marks each.

QUESTION 1 (30 Marks)

- A. A real estate developer is in the process of constructing a modern day vehicle gate barrier for his tenants in Kangemi Nyeri County. They have approached you seeking a solution that can enable them the gate to open only when the vehicle is registered in the estate and reject unregistered ones.
- When conducting needs analysis to this problem, except for this developer, who could be other customers to this problem? [2Marks]
 - List four questions you would investigate in this problem [2Marks]
 - Write a problem statement to this problem [2Marks]
 - Write two specifications applicable to this problem [2Marks]
 - Using neat sketches develop three alternative solutions to this problem and identify three criteria which may be used in selection phase [12 Marks]
- B. Paraplegic riders often requires to go back to the roads and ride again after being involved in an accident. You have been contracted as a consultant by ROAM, an electric bike manufacturing company to develop an electric bike for the paraplegic riders.
- What are the four main components of statement of work? [2 Marks]
 - Develop a statement of work (SOW) for this project [8 Marks]

SECTION A (ATTEMPT ONLY ONE QUESTION FROM THIS SECTION)

QUESTION 2 (20 Marks)

A successful hand tool manufacturer was exploring the growing market for handheld power tools. After performing initial research, the firm decided to enter the market with a cordless screwdriver.

- A. After conducting interview with one of the customers, the following listed customer statements were obtained. Write the interpreted need for each. [4Marks]
- I need to drive screws fast, faster than by hand.
 - I like the pistol grip; it feels the best.
 - I dont like it when the tip slips off the screw.
 - I would like to be able to lock it so I can use it with a dead battery.
- B. What are specifications and importance of specifications in product development process? For the screw driver problem write four sample metrics and corresponding specification statements [4 Marks]
- C. . What is the importance of clarifying the problem statement? Using hierarchy method clarify the cordless screw driver problem statement and choose the most appropriate problem statement. [7 Marks]

- D. . Carryout a functional decomposition of the screw driver problem [5 Marks]

QUESTION 3 (20 Marks)

- A. Wireless charging systems shown on **Figure Q3(A)** were designed by design team to allow the users to charge their electronics easily without the use of bulky cables

The team employed brain-writing creativity technique.

- i. Distinguish brain-writing from brain storming and clearly explain the process of brain-writing. [5 Marks]
- ii. Describe five different information generated during internal search indicating source and show how they may have influenced the design of the wireless charging systems. [5 Marks]



Figure Q3(A): Wireless charging system

- B. Briefly explain application of anthropometric data using following examples:-
- i. Controls [2 Marks]
 - ii. Work station designs [2 Marks]
 - iii. Biomechanics [2 Marks]
- C. The layout of an arcade video game console for use by 15-18 years old is as shown on **Figure Q 2(C)** with the popliteal height given . Identify four important areas on human/machine interface where the size would be important aspect of the design [4 Marks]

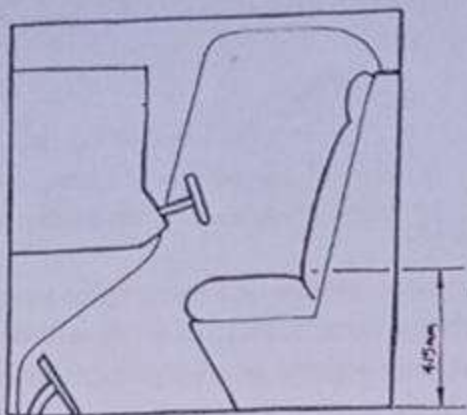


Figure Q 2(C): Video game Console

SECTION B (ATTEMPT ONE QUESTION FROM THIS SECTION)

QUESTION 4 (20 Marks)

- A. Explain any three ways through which ergonomic problems can be identified in a work place [6 marks]
- B. Dominic is a process controller in a large assembly plant. Much of his work involves using a computer (display screen equipment or DSE) for controlling a number of systems, record messages and respond to e-mails. Dominique regularly uses a telephone to take messages. After working at the control desk for eight months, Dominic found he was leaving work with an aching shoulder and neck, and with sore eyes and a headache. Dominic asked his manager for a risk assessment to identify possible problems with his working arrangements and to determine where improvements could be made.
 - i. State and explain any two areas for improvement which the manager could identify during the risk assessment. [4 marks]
 - ii. State any four simple, cost effective measures to reduce the risks identified. [4 marks]
- C. State one advantage and one disadvantage of each of the following control devices.
 - i. Bar Knob [2 Marks]
 - ii. Adjustable handle [2 Marks]
 - iii. Arm hand wheel [2 Marks]

QUESTION 5 (20 Marks)

- A. Describe the following terms as used in engineering design
 - i. Modeling [2 Marks]
 - ii. Simulation [2 Marks]

REB
24

- B. **Figure Q5 (B)** shows a machine-shop that has 2 lathe machines, one milling machine and a grinder. Two types of parts enter the machine shop. Type 1 parts require turning, milling and grinding in sequence. Type 2 parts require only milling and grinding. The frequency of arrival and the time to be routed to the turning area are deterministic for both types of parts. Use this as an example to illustrate the steps followed in simulation modeling [10 Marks]

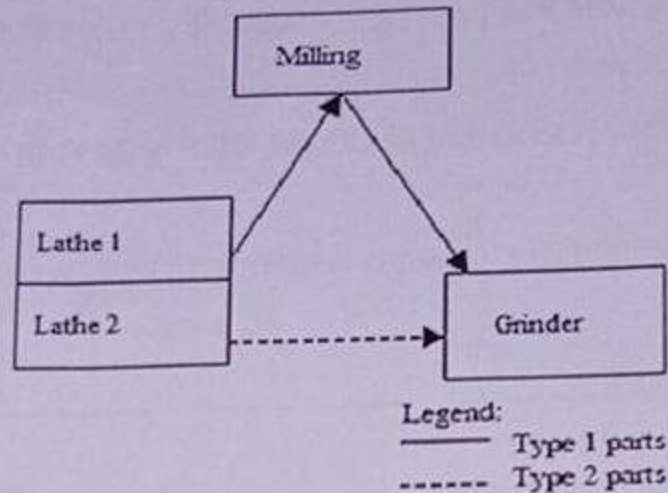


Figure Q5(B)

- C. **Figure Q5(C)** depicts the inside of a furniture maker's workshop. Evaluate this figure to identify ergonomic problems with the work activities and environment, and recommend solutions on how the owner of the workshop can improve them. (NB. Use a table to give the problems and the associated solutions.) [6 Marks]



Figure Q5(C): Furniture workshop